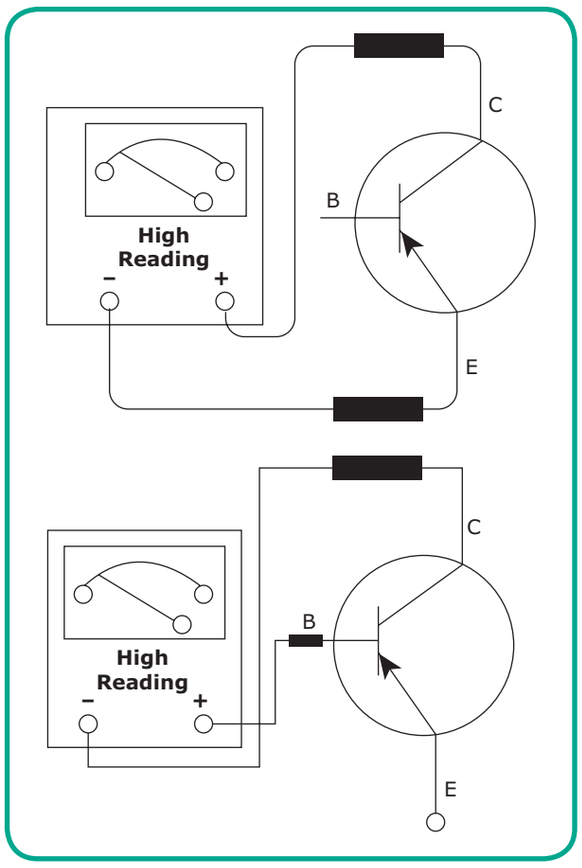


Figure 5.4 PNP and NPN transistor testing

5. Collector to Base
6. Base to Collector

Each pair should have one side with very high resistance (> 1 MΩ) and the other side with a much lower resistance of a few hundred thousand ohms. If this is the case for all the transistor leads, the transistor is good. If not, transistor is defective.

5.6. TRANSISTORS BIASING

Various Methods of Transistor Biasing

1. Base resister method or fixed bias method
2. Collector to base bias method or feedback bias method
3. Self-bias or emitter or voltage divider bias

Transistor biasing is the process of setting a transistor DC operating voltage or current condition to the required level. So that, any AC input signal can be amplified correctly by the transistor.

5.6.1 Fixed Bias

Fixed biasing circuit is very simple, biasing conditions can be easily set, calculations are very simple but there is no loading of source by the biasing circuit. Hence, this method is rarely used, due to poor stabilization. Figure 5.5 shows fixed biased method of a transistor.

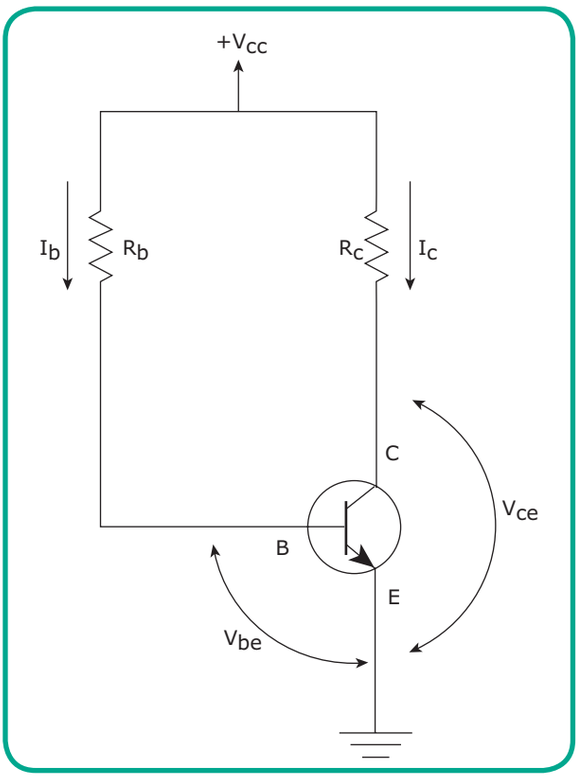


Figure 5.5 Fixed bias

5.6.2 Feedback Bias

This method is simple and provides better biasing stability but the circuit provides a negative feedback resulting in reduced amplifier gain and fairly high stability factor.

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